Half-spaces and polyhedra

Outline

Half-spaces and supporting hyperplanes

Polyhedra

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Half-spaces and supporting hyperplanes

Introduction: Half-spaces and supporting hyperplanes

- A single linear inequality $\sum_{j=1}^{n} a_j x_j \leq b$ splits $\mathbb{R}^n$ into two parts: the points that satisfy it form a **half-space**
- The points that satisfy it with equality form its **supporting hyperplane**
- Half-spaces are the building blocks of everything that follows: every constraint of a linear programming problem defines one
- We give the definitions and draw two half-spaces in $\mathbb{R}^2$

Half-space

Definition 1: half-space

Given a (row) vector

$$\underbrace{\begin{pmatrix} a_1 & a_2 & \dots & a_n \end{pmatrix}}_{\boldsymbol{a}} \in \mathbb{R}^{1 \times n}$$

and a scalar $b \in \mathbb{R}$*, the set*

$$\left\{ \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n : \underbrace{\sum_{j=1}^{n} a_j \, x_j \leq b}_{\boldsymbol{a}\,\boldsymbol{x} \leq b} \right\}$$

is called a ***half-space****.*

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Supporting hyperplane

- Given a half-space, the set

$$\left\{ \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n : \quad \underbrace{\sum_{j=1}^{n} a_j \, x_j = b}_{\boldsymbol{a}\,\boldsymbol{x}=b} \right\}$$

  is called the **supporting hyperplane** of the half-space.

- The inequality $\leq$ becomes an equality $=$: the hyperplane is the **boundary** of the half-space

Let us now draw two half-spaces in $\mathbb{R}^2$:

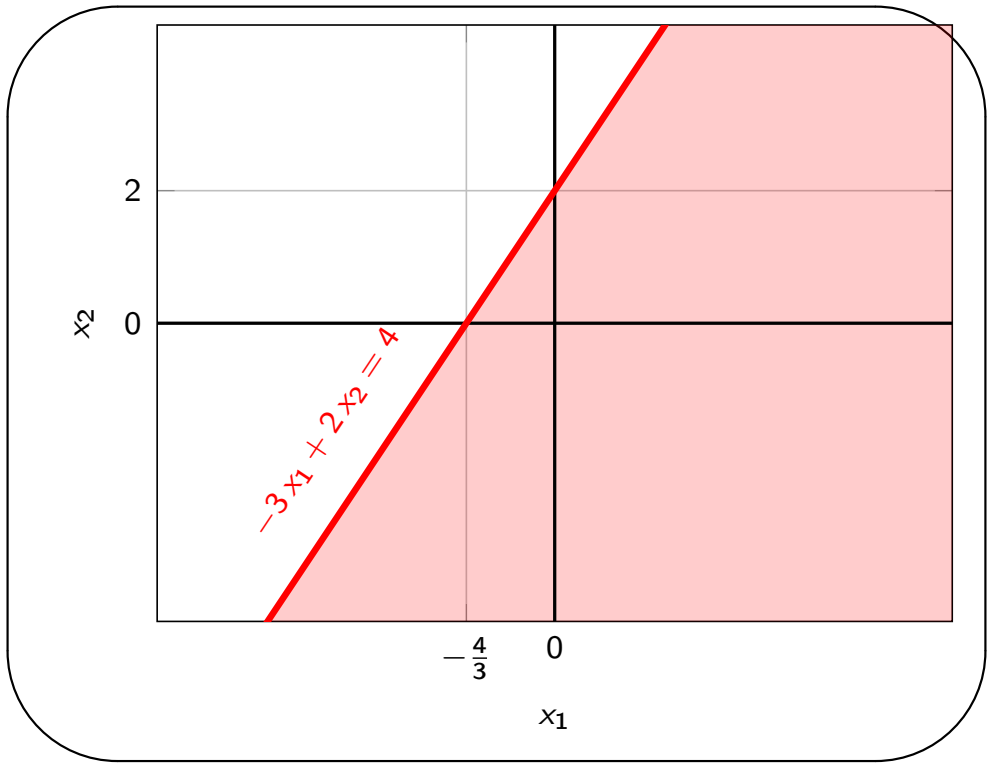
the shaded part is the half-space,
the red line is its supporting hyperplane.

Example (1/2)

Let us consider the following half-space in $\mathbb{R}^2$:

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \quad -3\,x_1 + 2\,x_2 \leq 4 \right\}$$

$$\text{where} \quad \boldsymbol{a} = \begin{pmatrix} -3 & 2 \end{pmatrix} \quad \text{and} \quad b = 4$$

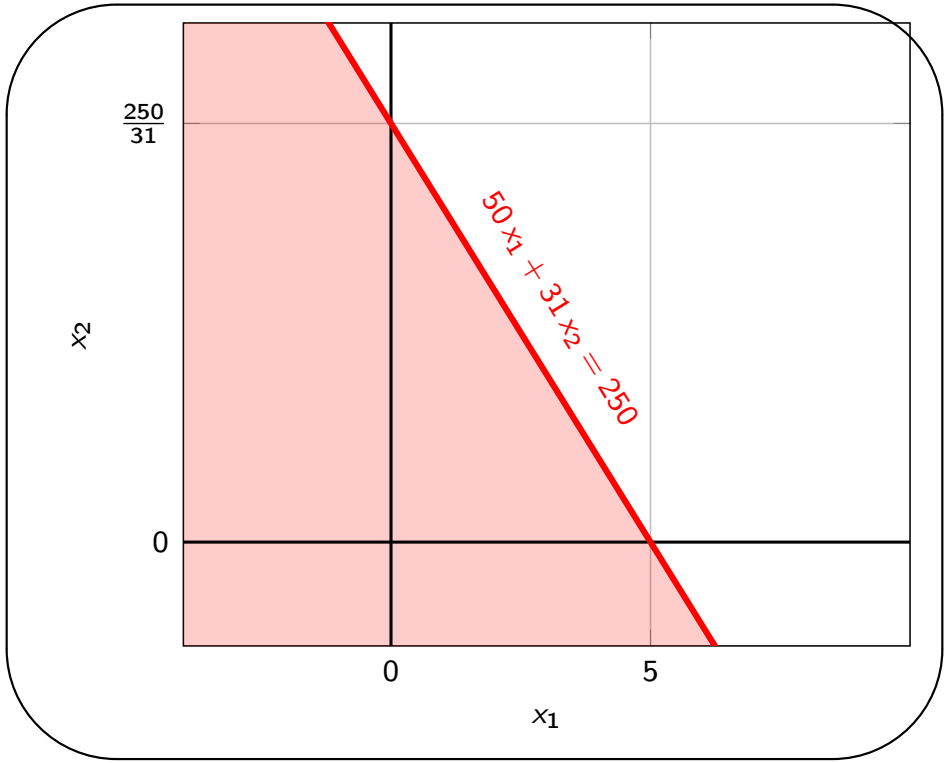


Example (2/2)

Let us consider the following half-space in $\mathbb{R}^2$:

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \quad 50\, x_1 + 31\, x_2 \leq 250 \right\}$$

$$\text{where} \quad \boldsymbol{a} = \begin{pmatrix} 50 & 31 \end{pmatrix} \quad \text{and} \quad b = 250$$



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Summary: half-spaces and supporting hyperplanes

Half-space:

- Given $\boldsymbol{a} \in \mathbb{R}^{1 \times n}$ and $b \in \mathbb{R}$, the set $\{\boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{a}\,\boldsymbol{x} \leq b\}$
- The points that satisfy one linear inequality

Supporting hyperplane:

- The set $\{\boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{a}\,\boldsymbol{x} = b\}$
- In $\mathbb{R}^2$ it is a line, and the half-space is one of the two sides of the line

Polyhedra

Introduction: Polyhedra

- A linear programming problem has **many** constraints, not just one
- The points that satisfy all of them at the same time form the intersection of several half-spaces: a **polyhedron**
- We define polyhedra in matrix notation, bounded polyhedra and polytopes
- We see that equality constraints (hyperplanes) are allowed too, with two examples in $\mathbb{R}^2$ and in $\mathbb{R}^3$

Polyhedron (1/2)

Definition 2: polyhedron

Given a matrix and a (column) vector

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}}_{\boldsymbol{A}} \in \mathbb{R}^{m \times n} \qquad \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\boldsymbol{b}} \in \mathbb{R}^{m}$$

Polyhedron (2/2)

Definition: polyhedron

. . . the set

$$\left\{ \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{R}^n : \quad \underbrace{\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\}}_{\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}} \right\}$$

is called a ***polyhedron****.*

- Each row $i \in \{1, 2, \ldots, m\}$ of $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$ is a half-space: a polyhedron is the **intersection of m half-spaces**
- Some polyhedra extend to infinity, others stay in a bounded region of $\mathbb{R}^n$
- Let us make this distinction precise

Definition 3: bounded polyhedron

*A polyhedron $P \subseteq \mathbb{R}^n$ is **bounded** if there exists*

$$\underbrace{\begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix}}_{=\boldsymbol{\beta}} \in \mathbb{R}^n_{\geq 0}$$

such that

$$\underbrace{-\beta_j \leq x_j \leq \beta_j}_{|x_j| \leq \beta_j}, \qquad \forall \boldsymbol{x} \in P,\ \forall j \in \{1, 2, \ldots, n\}.$$

*Here, β_j is an **upper bound**, while $-\beta_j$ is a **lower bound** of P along dimension $j \in \{1, 2, \ldots, n\}$.*

Polytope

Definition 4: polytope

A bounded polyhedron is called a ***polytope****.*

Let us look at a polyhedron in $\mathbb{R}^2$
defined by four half-spaces,

and check that it is a polytope.

Example (1/4)

Consider the following polyhedron defined as the intersection of four half-spaces:

$$P = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \begin{array}{r} -x_1 - x_2 \leq -3 \\ x_1 + x_2 \leq 9 \\ -x_1 + x_2 \leq 3 \\ x_1 - x_2 \leq 3 \end{array} \right\}$$

- Each row is one half-space: $m = 4$ and $n = 2$

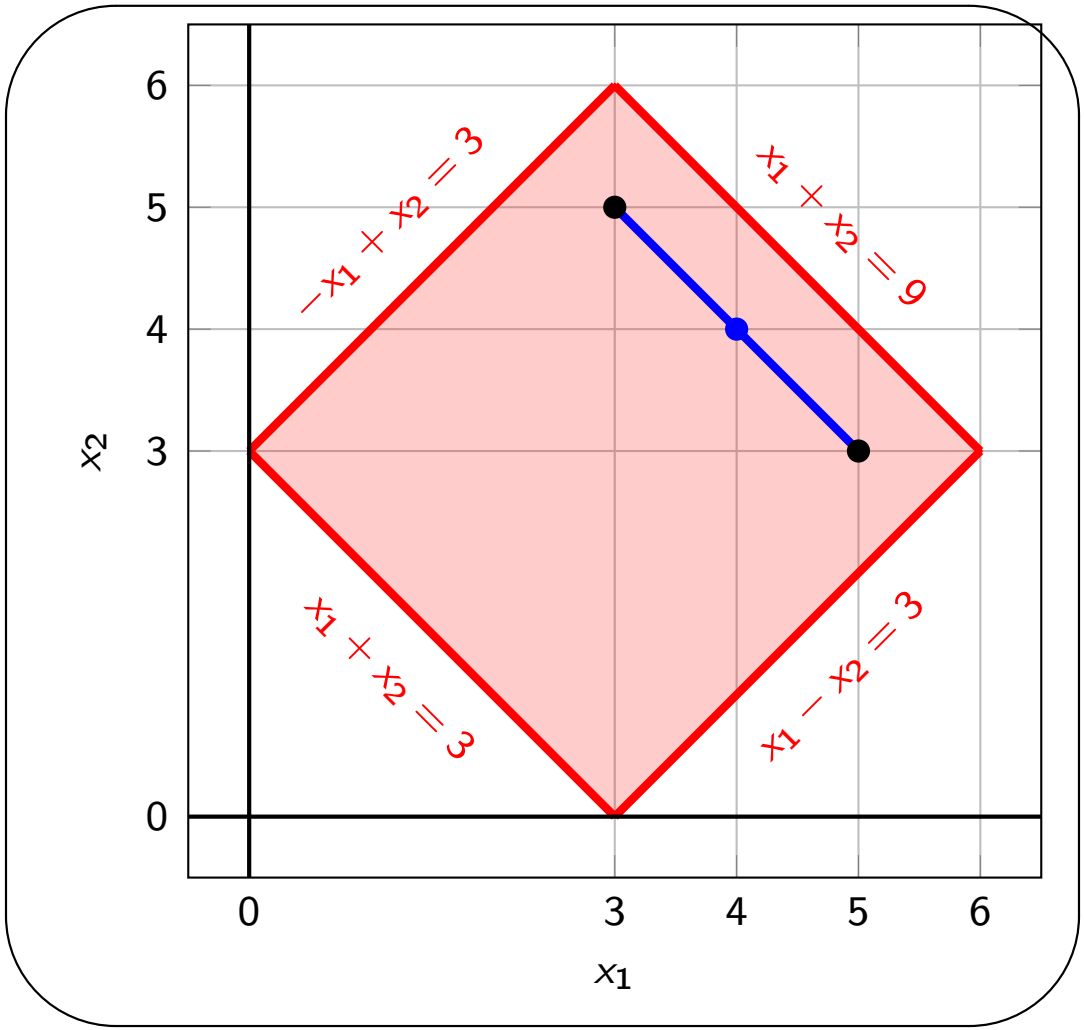
Example (2/4)

- Consider two points, $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$, in the set P
- All points on the blue segment in the figure, given by:

$$\lambda \begin{pmatrix} 3 \\ 5 \end{pmatrix} + (1-\lambda) \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5-2\lambda \\ 3+2\lambda \end{pmatrix} \quad \text{for} \quad \lambda \in [0,1],$$

  also belong to P
- Indeed, for every $\lambda \in [0,1]$ we have $x_1 + x_2 = 8$ and $x_1 - x_2 = 2 - 4\lambda \in [-2,2]$, so that all four inequalities defining P are satisfied

Example (3/4)



Example (4/4)

- For example, with $\lambda = \frac{1}{2}$, the point of the segment is the one shown in blue

$$\begin{pmatrix} \frac{1}{2}3 + (1 - \frac{1}{2})5 \\ \frac{1}{2}5 + (1 - \frac{1}{2})3 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

- This polyhedron P is also a **polytope**, since it is a bounded polyhedron in $\mathbb{R}^2$
- Indeed, by setting $\beta = \begin{pmatrix} 6 \\ 6 \end{pmatrix}$ we have

$$|x_1| \leq 6 \quad \text{and} \quad |x_2| \leq 6 \quad \text{for all} \quad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in P$$

Half-spaces and hyperplanes

- It is worth noticing that if a polyhedron contains the following two half-spaces:

$$\{\boldsymbol{x} \in \mathbb{R}^n : \quad \boldsymbol{a}\,\boldsymbol{x} \leq b\} \qquad \text{and} \qquad \{\boldsymbol{x} \in \mathbb{R}^n : \quad \boldsymbol{a}\,\boldsymbol{x} \geq b\}$$

it contains, *de facto*, the following hyperplane:

$$\{\boldsymbol{x} \in \mathbb{R}^n : \quad \boldsymbol{a}\,\boldsymbol{x} = b\}$$

In general, a polyhedron is given by the intersection of **half-spaces** and **hyperplanes**.

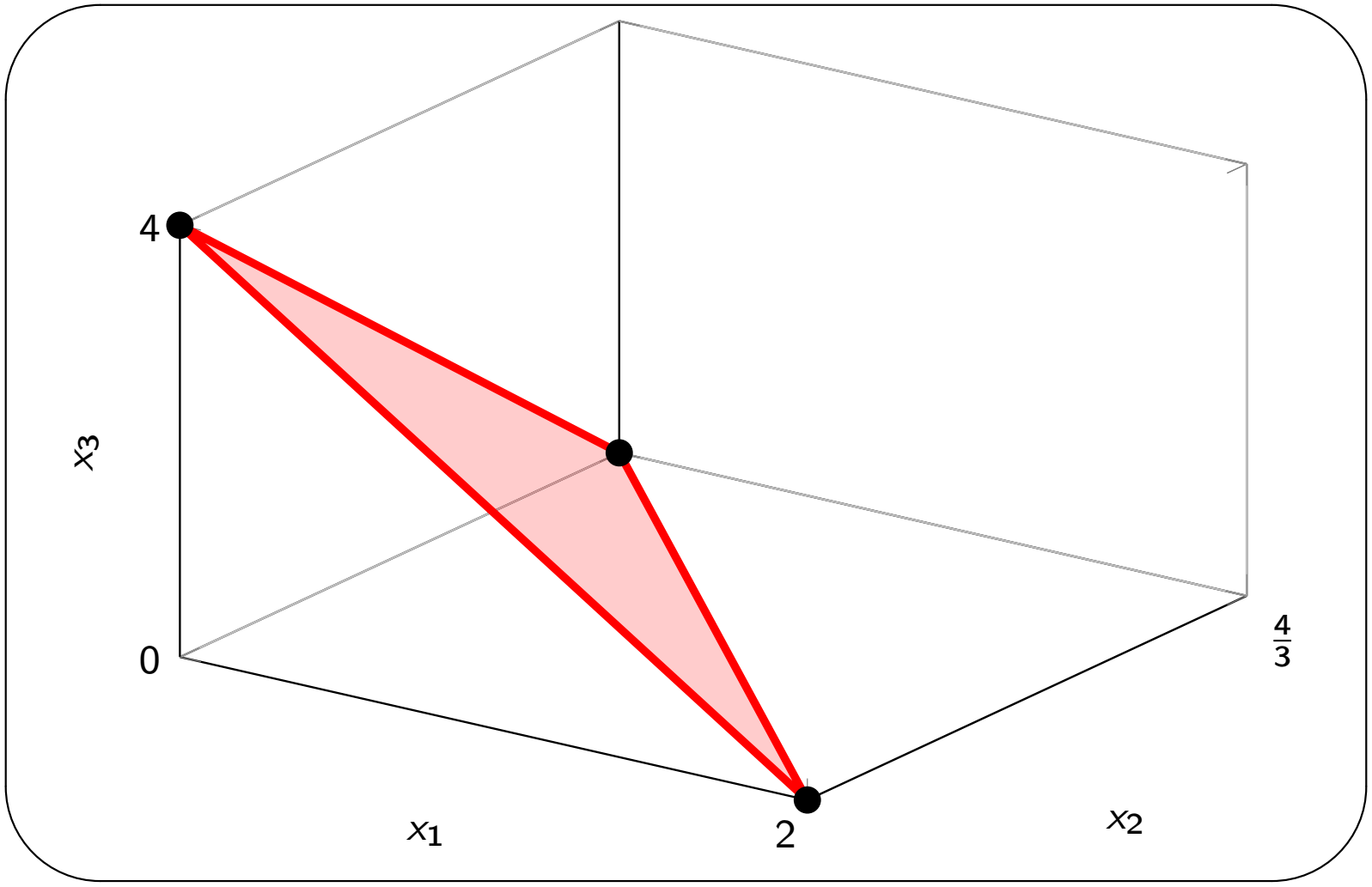
Example (1/2)

Consider the following polyhedron given by the intersection of three half-spaces and one hyperplane:

$$P = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{aligned} x_1 &\geq 0 \\ x_2 &\geq 0 \\ x_3 &\geq 0 \\ 2x_1 + 3x_2 + x_3 &= 4 \end{aligned} \right\}$$

- Its three-dimensional representation corresponds to the hyperplane shown in the figure on the next slide

Example (2/2)



Summary: polyhedra

Polyhedron:

- The set $\{\boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}\}$, with $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$: the intersection of m half-spaces
- In general, the intersection of half-spaces and hyperplanes (an equality is a pair of inequalities $\leq$ and $\geq$)

Bounded polyhedron and polytope:

- Bounded: $|x_j| \leq \beta_j$ for all $\boldsymbol{x} \in P$ and all $j \in \{1, 2, \ldots, n\}$
- A bounded polyhedron is a polytope

Summary

- A **half-space** is the set of points that satisfy one linear inequality $\boldsymbol{a}\,\boldsymbol{x} \leq b$; its **supporting hyperplane** is the set where $\boldsymbol{a}\,\boldsymbol{x} = b$
- A **polyhedron** is the set $\{\boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}\}$: the intersection of half-spaces and, in general, of hyperplanes
- A polyhedron is **bounded** if every variable has a lower bound $-\beta_j$ and an upper bound β_j on it
- A bounded polyhedron is a **polytope**, like the polyhedron of the example in $\mathbb{R}^2$